Yaser Husen N

AI/ML Engineer | LLM Applications, RAG and Agentic Systems | Python

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PROFESSIONAL SUMMARY

AI/ML engineer specialising in LLM-powered applications, retrieval-augmented generation (RAG), and multi-agent systems, with Python as the primary language. Designs, builds, evaluates and deploys AI systems end to end: model fine-tuning and quantization, FastAPI backends, vector search, and the interfaces that deliver them to users. Has built and deployed original AI systems (AEEA, ATTF), holds Google Cloud Vertex AI certifications, and has delivered production client projects since 2023. Directs and reviews AI-assisted development; learns quickly, takes ownership and ships working software.

AREAS OF EXPERTISE

Large Language Model (LLM) applications | Retrieval-Augmented Generation (RAG) | AI agents and multi-agent orchestration | Prompt engineering | Model fine-tuning (LoRA) and quantization | Model evaluation | Python backend and API development | Local and offline AI deployment | Speech (STT/TTS) pipelines

TECHNICAL SKILLS

Python	FastAPI, async SQLAlchemy, Celery, Pydantic, REST API design, scripting and automation, LLM API clients (Gemini, Claude), automated testing
AI / ML	LLMs, RAG, vector search (FAISS), prompt engineering (few-shot, chain-of-thought, structured JSON output), LoRA fine-tuning (Unsloth), GGUF quantization, Ollama, llama.cpp, model evaluation and benchmarking
Speech AI	Moonshine and Whisper.cpp (speech-to-text), Kokoro (text-to-speech), low-latency voice pipelines
Cloud and MLOps	Google Cloud Platform (Vertex AI, Cloud Run), Docker, Docker Compose, Redis, Git, Vercel, CI-style validation suites
Other Languages	JavaScript, Java, SQL, C, C++
Web	React, Next.js, Node.js, Express, MongoDB, HTML, CSS

PROJECTS

AEEA: Adaptive Entropy-Equilibrium Architecture

Original AI research framework (Python, React, Spring Boot) | *Live app* | *Simulator*

- Designed an 8-stage framework that models AI training as a manufacturing pipeline, documented in a 20-section blueprint refined through a formal audit that resolved 8 issue categories.
- Architected multi-model orchestration in which a Claude meta-controller routes tasks across Gemini, Claude and local Ollama models through a Redis Streams arbitration bus.
- Delivered an interactive React simulator (8 tabs, 30 of 32 automated validation checks passing) and a Spring Boot web app with AES-256-GCM encrypted API keys, OTP password reset and role-based access control.

ATTF: Adaptive Tit-for-Tat Framework

Game-theoretic AutoML pipeline (Python, FastAPI, Next.js) | *Live app*

- Built a data-quality pipeline in which preprocessing agents earn reputation from downstream model performance, so higher-quality data is selected automatically.
- Engineered the Python backend with FastAPI, async SQLAlchemy, Celery with Redis, and FAISS; modelled 14 ORM entities with async repositories and Pydantic schemas.
- Containerized the full stack with Docker Compose for CPU-only, self-hosted deployment.

Local LLM Fine-Tuning and Optimization

CPU-only environment (Python, Unsloth, llama.cpp, Ollama)

- Fine-tuned DeepSeek-R1 1.5B with LoRA (Unsloth) and deployed Qwen3 1.7B through llama.cpp.
- Benchmarked GGUF quantization levels and selected Q5_K_M (about 1.5 GB) after Q8_0 proved too slow, balancing output quality against latency.
- Wrote a 13-section engineering guide covering parameter tuning, Python API scripting, multi-turn memory and agent pipelines.

Nyx: Privacy-First Local AI Agent
In development (Python)

- Building a fully offline desktop agent with a local LLM, IDE copilot, tool use, and voice conversation (Moonshine STT, Kokoro TTS), plus a staged self-improvement engine with snapshots and rollback.
- Designed for low-latency voice interaction and zero runtime data egress; verified with an automated test suite.

RAG Chatbot on Google Cloud
Vertex AI

- Built a retrieval-augmented generation chatbot on Vertex AI for enterprise knowledge retrieval.

Gemini Sentiment Analyzer and Behavior Evaluator
LLM API projects

- Engineered few-shot and chain-of-thought prompts that return structured JSON sentiment scores, rendered live in a web interface.
- Built a multi-turn Claude and Gemini pipeline that generates structured SWOT reports, reducing generation time from hours to under 10 seconds.

PROFESSIONAL EXPERIENCE

Front-End Developer and Client-Side Consultant
Aspira | 2023 to Present

- Delivered production websites from design through deployment for a gym business and local vendors.
- Integrated third-party APIs and LLM features, including AI-generated FAQ and content sections, into client products, reducing client support queries.
- Translated client requirements into clear technical scope up front, reducing revision cycles.

EDUCATION

Bachelor of Science, Computer Science
Islamiah College (Thiruvalluvar University) | Expected 2027
Relevant coursework: Data Structures, Cloud Computing, Database Management, UI/UX Design

CERTIFICATIONS AND ACHIEVEMENTS

- Google Cloud (Nov 2025): Explore Generative AI with the Vertex AI Gemini API; Prompt Design in Vertex AI; Gen AI Agent: Transforming Your Organisation
- Python for Data Science, TechAcademy (Jun 2026)
- freeCodeCamp: Responsive Web Design; JavaScript Algorithms and Data Structures
- NPTEL (Swayam): Computer Science fundamentals
- Winner, "AI Integration in Society" presentation competition
- Conductor, DREAM IT '26 Technical Symposium

LANGUAGES

English, Tamil, Hindi (native); Urdu (conversational)